DENOMINATORS OF CURTZ'S POLYNOMIALS ARE FARHI'S $d(n+1,k+1)$: A FORMAL PROOF OF MARCUS'S CONJECTURE FOR OEIS A141412

MATH0

ABSTRACT. Curtz's polynomials are defined by $P_0(x) = 1$ and $P_n(x) = (-1)^n/(n+1) + x\sum_{i=0}^{n-1} \frac{(-1)^i}{i+1} P_{n-1-i}(x)$ for $n \geq 1$, and OEIS A141412 is the triangle $T(n,k)$ of the reduced denominators of the coefficients $[x^k]P_n(x)$. In 2018 Michel Marcus conjectured in that entry that $T(n,k) = d(n+1,k+1)$, where $d(n,k)$ is the denominator of $\left[{n \atop k}\right]k!/n!$, with $\left[{n \atop k}\right]$ the unsigned Stirling numbers of the first kind; these are the numbers $d_{n,k}$ of Farhi. We give a machine-checked proof of this conjecture for every entry $0 \leq k \leq n$ of the triangle. The proof goes through the identity $[x^k]P_n(x) = (-1)^{n+k}(k+1)!\left[{n+1 \atop k+1}\right]/(n+1)!$ $(n,k \geq 0)$, from which the conjecture follows at once and which is checked in Lean as an auxiliary lemma. This identity is not new: it is equation (6) of a 1993 paper of Flajolet, Gourdon and Salvy, and it is also recorded in OEIS formulas of Curtz (2008) and Greubel (2023); our contribution is the first formal proof of this known result. The proof is verified in Lean 4 with Mathlib against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before any proof attempt; its definitions are checked by the kernel against all 78 listed terms.

1. INTRODUCTION

In 1992 Paul Curtz [2] introduced the polynomials $P_n(x) \in \mathbb{Q}[x]$ defined by

$$P_0(x) = 1, \qquad P_n(x) = \frac{(-1)^n}{n+1} + x\sum_{i=0}^{n-1} \frac{(-1)^i}{i+1}\, P_{n-1-i}(x) \quad (n \geq 1). \tag{1}$$

They are the characteristic polynomials of the matrices that occur in Adams' method for the numerical solution of ordinary differential equations [5], and Curtz conjectured in [2] that P_n has no real root for even n and exactly one positive real root for odd n. Flajolet, Gourdon and Salvy [5] proved this conjecture in 1993. Their starting point is the bivariate generating function

$$\sum_{n\geq 0} P_n(x) z^n = \frac{\log(1+z)}{z} \cdot \frac{1}{1 - x\log(1+z)} \tag{2}$$

([5, eq. (3)]), which they compare with the classical generating function of the Stirling numbers of the first kind [1] to obtain ([5, eq. (6)])

$$P_n(x) = \frac{(-1)^n}{(n+1)!} \sum_{k=0}^{n} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} (k+1)!\,(-x)^k, \tag{3}$$

where $\left[{n \atop k}\right]$ denotes the unsigned Stirling number of the first kind (the number of permutations of n elements with k cycles).

In the OEIS the polynomials are defined in a comment of A129891 [10] (whose terms are the row sums of a related family of polynomials with integer coefficients, not of the P_n), and the numerators and denominators of their coefficients $c(n,k) = [x^k]P_n(x)$ form the triangles A140749 [12] and A141412 [13]; all three entries are due to Curtz. A141412 lists the triangle $T(n,k) = \operatorname{den} c(n,k)$ for $0 \leq k \leq n \leq 11$. On October 18, 2018, Michel Marcus added to

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A141412 the conjecture that $T(n,k) = d(n+1,k+1)$, where $d(n,k)$ is the denominator of $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right] k!/n!$, referring to the numbers $d_{n,k}$ of Farhi [3]. Farhi defines $F_{n,k} = \sum 1/(i_1 i_2 \cdots i_k)$, the sum over all $(i_1, \ldots, i_k)$ of positive integers with $i_1 + \cdots + i_k = n$, and $d_{n,k} = \operatorname{den} F_{n,k}$; by [3, Prop. 4.1], $F_{n,k} = \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right] k!/n!$, so Marcus's $d(n,k)$ is Farhi's $d_{n,k}$.

Status and prior work. Formula (3) gives $c(n,k) = (-1)^{n+k}\left[\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix}\right](k+1)!/(n+1)!$, and since a sign does not change a reduced denominator, Marcus's conjecture follows immediately from the 1993 paper [5], which the entry A141412 itself cites. The OEIS also records the numerator and denominator parts of this identity, in terms of the triangle $A048594(n,k) = k!\, s(n,k)$ of signed Stirling numbers, whose k-th column has exponential generating function $\log(1+x)^k$ [9]: in A140749 Curtz noted on July 17, 2008 that the numerators of the $c(n,k)$ are those of $A048594(n+1,k+1)/(n+1)!$, and in A141412 G. C. Greubel added on October 24, 2023 the formula "Equals denominators of $A048594(n+1,k+1)/(n+1)!$", with Mathematica, Magma and Sage programs that compute the triangle this way. Nevertheless the entry (revision #32, November 7, 2023) still labels Marcus's statement a conjecture.

The mathematics in this paper is therefore known; the result is a *first formal proof of a known result*. The math0 referees found no earlier formal proof of the statement, and it is not among the OEIS statements of the formal-conjectures repository [4] (at the commit pinned in Section 4). Our proof (Section 3) is the column-by-column form of the generating-function argument of [5], with the Stirling generating function derived from a differential equation.

2. Statement

Definition 1. Let $P_n(x) \in \mathbb{Q}[x]$ be defined by (1) and let $c(n,k) = [x^k]P_n(x)$ for $n,k \geq 0$. For $q \in \mathbb{Q}$ let $\operatorname{den} q \geq 1$ be the denominator of q in lowest terms. Put

$$T(n,k) = \operatorname{den} c(n,k), \qquad d(n,k) = \operatorname{den}\Big(\begin{bmatrix} n \\ k \end{bmatrix} \frac{k!}{n!}\Big) \qquad (n,k \geq 0),$$

where $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$ is the unsigned Stirling number of the first kind, given by $\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right] = 1$, $\left[\begin{smallmatrix} 0 \\ k+1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} n+1 \\ 0 \end{smallmatrix}\right] = 0$ and $\left[\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix}\right] = n\left[\begin{smallmatrix} n \\ k+1 \end{smallmatrix}\right] + \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$.

The first rows of $(c(n,k))_{0 \leq k \leq n}$ are

$$1; \quad -\tfrac{1}{2},\ 1; \quad \tfrac{1}{3},\ -1,\ 1; \quad -\tfrac{1}{4},\ \tfrac{11}{12},\ -\tfrac{3}{2},\ 1; \quad \tfrac{1}{5},\ -\tfrac{5}{6},\ \tfrac{7}{4},\ -2,\ 1; \quad \ldots,$$

so that T begins $1;\ 2,1;\ 3,1,1;\ 4,12,2,1;\ 5,6,4,1,1$, as in A141412. The FORMULA section of A141412 [13] contains, verbatim:

> `Conjecture: T(n, k) = d(n+1, k+1), with d(n,k) = denominator(A000254(n, k)*k!/n!) where A000254 are the unsigned Stirling numbers of the 1st kind. See d(n,k) in Farhi link. - _Michel Marcus_, Oct 18 2018`

Strictly, A000254 [8] is the one-index sequence $\left[\begin{smallmatrix} n+1 \\ 2 \end{smallmatrix}\right]$; the comment uses it as a function of two arguments and states that it means the unsigned Stirling numbers of the first kind, so we read $A000254(n,k)$ as $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$ (the triangle A132393 [11]). This agrees with Farhi's $d_{n,k}$ and with Greubel's formula and programs in the entry.

Theorem 2 (Marcus's conjecture)**.** *For all $n \geq 0$ and $0 \leq k \leq n$, $T(n,k) = d(n+1,k+1)$.*

Theorem 2 follows from the following identity, which is (3).

Theorem 3 ([5, eq. (6)])**.** *For all $n,k \geq 0$,*

$$c(n,k) = (-1)^{n+k} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} \frac{(k+1)!}{(n+1)!}.$$

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A141412.lean`, reviewed for fidelity and frozen (SHA-256 `3f92df65...7b77`) before any proof attempt. Its definitions read, verbatim (header, import, docstrings, blank lines and the three theorems are omitted):

```
namespace Math0.Statements.A141412
open Polynomial
noncomputable def P : ℕ → ℚ[X]
  | 0 => 1
  | n + 1 => C ((-1) ^ (n + 1) / (((n + 1 : ℕ) : ℚ) + 1)) +
      X * ∑ i ∈ Finset.range (n + 1), C ((-1) ^ i / ((i : ℚ) + 1)) * P (n - i)
decreasing_by omega
def c : ℕ → ℕ → ℚ
  | 0, 0 => 1
  | 0, _ + 1 => 0
  | n + 1, 0 => (-1) ^ (n + 1) / (((n + 1 : ℕ) : ℚ) + 1)
  | n + 1, k + 1 => ∑ i ∈ Finset.range (n + 1), (-1) ^ i / ((i : ℚ) + 1) * c (n - i) k
def T (n k : ℕ) : ℕ := (c n k).den
def d (n k : ℕ) : ℕ :=
  ((Nat.stirlingFirst n k : ℚ) * (Nat.factorial k : ℚ) / (Nat.factorial n : ℚ)).den
def statement : Prop :=
  ∀ n k : ℕ, k ≤ n → T n k = d (n + 1) (k + 1)
end Math0.Statements.A141412
```

The encoding matches Definition 1 as follows. The polynomial `P` is (1) written for $n+1$: $P_{n+1} = (-1)^{n+1}/(n+2) + X\sum_{i=0}^{n} \frac{(-1)^i}{i+1} P_{n-i}$. Mathlib polynomials are noncomputable, so the table entries are given by the computable recursion `c`, which is the coefficientwise form of the same recursion; the theorem `coeff_P` in the same file, `(P n).coeff k = c n k` for all `n k`, proves by induction on k that the two agree. `T` takes `Rat.den`, the positive denominator of a rational number in lowest terms, as Maple's `denom` in the entry's program; the theorem `T_eq_den_coeff`, `T n k = ((P n).coeff k).den`, restates $T(n,k) = \mathrm{den}([x^k]P_n)$. `T_data` states that the list of the 78 values $T(n,k)$, $0 \le k \le n \le 11$, read by rows (written `(List.range 12).flatMap (fun n => (List.range (n + 1)).map (fun k => T n k))`), equals the list of the terms in A141412; it is proved by `decide +kernel`, i.e. by evaluation in the Lean kernel, in about 9 seconds. `Nat.stirlingFirst` is Mathlib's unsigned Stirling number of the first kind, defined by exactly the recursion of Definition 1; in `d` the division is in $\mathbb{Q}$. The statement quantifies over the whole triangle $0 \le k \le n$ and adds no hypotheses. The proof does not use the hypothesis $k \le n$; for $k > n$ both sides equal 1.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A141412.lean`; the name of the corresponding Lean declaration is given with each step (all of them are `private`, except `result`). We work in the ring $\mathbb{Q}[\![X]\!]$ of formal power series, and write $[X^N]g$ for the coefficient of X^N in g and g' for the formal derivative. Define

$$L = \sum_{j\ge 1} \frac{(-1)^{j+1}}{j} X^j \qquad \text{(the series } \log(1+X)\text{; } \texttt{L}\text{)},$$

$$F_m = \sum_{N\ge 0} (-1)^{N+m} \begin{bmatrix} N \\ m \end{bmatrix} \frac{m!}{N!} X^N \qquad (m \ge 0\text{; } \texttt{F}),$$

$$C_k = \sum_{n\ge 0} c(n,k) X^n \qquad (k \ge 0\text{; } \texttt{Cs}).$$

Thus $[X^0]L = 0$ and $[X^j]L = (-1)^{j+1}/j$ for $j \geq 1$.

Lemma 4 (`coeff_one_add_X_mul_deriv`)**.** *For every $g \in \mathbb{Q}[\![X]\!]$ and $N \geq 0$, $[X^N]\big((1+X)g'\big) = (N+1)[X^{N+1}]g + N[X^N]g$.*

Proof. We have $[X^N]g' = (N+1)[X^{N+1}]g$. For $N \geq 1$, $[X^N](Xg') = [X^{N-1}]g' = N[X^N]g$; for $N = 0$, $[X^0](Xg') = 0 = N[X^N]g$. Add the two. □

Lemma 5 (`one_add_X_mul_deriv_L`)**.** $(1+X)L' = 1$.

Proof. By Lemma 4, the constant coefficient is $1 \cdot [X^1]L + 0 = 1$, and for $N \geq 1$ the coefficient of X^N is $(N+1)\cdot(-1)^{N+2}/(N+1) + N \cdot (-1)^{N+1}/N = (-1)^N - (-1)^N = 0$. □

Lemma 6 (`one_add_X_mul_deriv_F`)**.** *For every $m \geq 0$, $(1+X)F'_{m+1} = (m+1)F_m$.*

Proof. Fix $N \geq 0$ and put $\varepsilon = (-1)^{N+m}$. By Lemma 4, the Stirling recurrence $\left[{N+1 \atop m+1}\right] = N\left[{N \atop m+1}\right] + \left[{N \atop m}\right]$ and $(N+1)! = (N+1)N!$,

$$\begin{aligned}[X^N]\big((1+X)F'_{m+1}\big) &= (N+1)\,\varepsilon \begin{bmatrix} N+1 \\ m+1 \end{bmatrix} \frac{(m+1)!}{(N+1)!} + N\,(-\varepsilon) \begin{bmatrix} N \\ m+1 \end{bmatrix} \frac{(m+1)!}{N!} \\ &= \varepsilon \frac{(m+1)!}{N!}\Big(N \begin{bmatrix} N \\ m+1 \end{bmatrix} + \begin{bmatrix} N \\ m \end{bmatrix} - N \begin{bmatrix} N \\ m+1 \end{bmatrix}\Big) = (m+1)\,\varepsilon \begin{bmatrix} N \\ m \end{bmatrix} \frac{m!}{N!},\end{aligned}$$

which is $[X^N]\big((m+1)F_m\big)$, since $(m+1)! = (m+1)m!$. In Lean: `Nat.stirlingFirst_succ_succ`, `Nat.factorial_succ`, then `field_simp` and `ring`. □

Lemma 7 (`F_eq`)**.** *For every $m \geq 0$, $F_m = L^m$.*

Proof. By induction on m. Since $\left[{N \atop 0}\right] = 1$ for $N = 0$ and 0 otherwise, $F_0 = 1 = L^0$. Assume $F_m = L^m$. By the power rule (`derivative_pow`) and Lemma 5,

$$(1+X)\big(L^{m+1}\big)' = (m+1)L^m \cdot (1+X)L' = (m+1)L^m = (m+1)F_m = (1+X)F'_{m+1},$$

the last step by Lemma 6. The constant coefficient of $1+X$ is 1, so $1+X \neq 0$, and $\mathbb{Q}[\![X]\!]$ is an integral domain; cancelling gives $(L^{m+1})' = F'_{m+1}$. The constant coefficients agree: $[X^0]F_{m+1} = \left[{0 \atop m+1}\right] \cdot (\dots) = 0$ and $[X^0]L^{m+1} = ([X^0]L)^{m+1} = 0$. Over a ring without additive torsion, such as $\mathbb{Q}$, a power series is determined by its derivative and its constant coefficient (Mathlib's `PowerSeries.derivative.ext`). Hence $F_{m+1} = L^{m+1}$. □

Lemma 7 is the classical identity $\sum_N s(N,m)z^N/N! = \log(1+z)^m/m!$ for the signed Stirling numbers $s(N,m) = (-1)^{N+m}\left[{N \atop m}\right]$ [1], which Mathlib does not provide.

Lemma 8 (`c_zero`)**.** *For every $n \geq 0$, $c(n,0) = (-1)^n/(n+1)$.*

Proof. For $n = 0$ both sides are 1; for $n \geq 1$ this is the defining equation of `c`. □

Lemma 9 (`Cs_succ`)**.** *For every $k \geq 0$, $C_{k+1} = L\,C_k$.*

Proof. The constant coefficients are $c(0,k+1) = 0$ and $[X^0]L \cdot c(0,k) = 0$. For $n \geq 0$,

$$[X^{n+1}](L\,C_k) = \sum_{j=0}^{n+1} [X^j]L\; c(n+1-j,k) = \sum_{i=0}^{n} [X^{i+1}]L\; c(n-i,k) = \sum_{i=0}^{n} \frac{(-1)^i}{i+1}\, c(n-i,k),$$

where we dropped the term $j = 0$ (as $[X^0]L = 0$) and put $j = i+1$; the right-hand side is the definition of $c(n+1,k+1)$. In Lean the reindexing uses `Finset.Nat.sum_antidiagonal_succ` and `Finset.Nat.sum_antidiagonal_eq_sum_range_succ_mk`. □

Lemma 10 (`X_mul_Cs`)**.** *For every $k \geq 0$, $X\,C_k = L^{k+1}$.*

Proof. By induction on k. For $k = 0$: both sides have constant coefficient 0, and for $n \geq 0$, $[X^{n+1}](XC_0) = c(n,0) = (-1)^n/(n+1) = (-1)^{n+2}/(n+1) = [X^{n+1}]L$ by Lemma 8. If $XC_k = L^{k+1}$, then $XC_{k+1} = X\,L\,C_k = L \cdot L^{k+1} = L^{k+2}$ by Lemma 9. □

Proof of Theorem 3 (`c_eq`). Comparing the coefficients of X^{n+1} in Lemma 10 and using Lemma 7 with $m = k+1$,

$$c(n,k) = [X^{n+1}](XC_k) = [X^{n+1}]L^{k+1} = [X^{n+1}]F_{k+1} = (-1)^{(n+1)+(k+1)} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix} \frac{(k+1)!}{(n+1)!},$$

and $(-1)^{(n+1)+(k+1)} = (-1)^{n+k}$. □

Proof of Theorem 2 (`result`). Let $n, k \geq 0$ and $q = \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}(k+1)!/(n+1)!$. By Theorem 3, $c(n,k)$ is q or $-q$ (`neg_one_pow_eq_or`). Since $\operatorname{den}(-q) = \operatorname{den} q$ (`Rat.neg_den`), $T(n,k) = \operatorname{den} c(n,k) = \operatorname{den} q = d(n+1,k+1)$. The hypothesis $k \leq n$ is not used. □

Remark 11 (not formalized)**.** Lemma 10 also yields Farhi's form of the coefficients directly. Expanding the power,

$$c(n,k) = [X^{n+1}]L^{k+1} = \sum_{\substack{i_1+\cdots+i_{k+1}=n+1 \\ i_j \geq 1}} \prod_{j=1}^{k+1} \frac{(-1)^{i_j+1}}{i_j} = (-1)^{n+k} F_{n+1,k+1},$$

with $F_{n,k}$ as in the introduction. Together with $F_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix} k!/n!$ [3, Prop. 4.1] this gives Theorem 3 again, and it shows that $T(n,k) = d_{n+1,k+1}$ holds with Farhi's original definition of $d_{n,k}$. Summing Lemma 10 over k with weight x^k gives (2).

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A141412.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A141412.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A141412.result`, of type `Math0.Statements.A141412.statement`. The auxiliary declarations are the private definitions and lemmas named in Section 3. The three series and the end of the file read, verbatim (header, imports, docstrings, blank lines, the `namespace` and `open` lines and the other lemmas omitted):

```
private noncomputable def L : ℚ⟦X⟧ :=
  PowerSeries.mk fun n => if n = 0 then 0 else (-1 : ℚ) ^ (n + 1) / n
private noncomputable def F (m : ℕ) : ℚ⟦X⟧ :=
  PowerSeries.mk fun N =>
    (-1 : ℚ) ^ (N + m) * (Nat.stirlingFirst N m : ℚ) * (m.factorial : ℚ) / (N.factorial : ℚ)
private noncomputable def Cs (k : ℕ) : ℚ⟦X⟧ :=
  PowerSeries.mk fun n => c n k
theorem result : statement := by
  intro n k _
  rw [T, d, c_eq]
  rcases neg_one_pow_eq_or ℚ (n + 1 + (k + 1)) with h | h
  · rw [h, one_mul]
  · rw [h, neg_one_mul, Rat.neg_den]
end Math0.Frontier.A141412
#math0_settles Math0.Frontier.A141412.result Math0.Statements.A141412.statement
```

The proof file uses no `sorry`, `decide`, `native_decide`, custom axioms or `set_option`; it does not use `T_data`, `P` or `coeff_P`, since `statement` is expressed through `c` directly.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or ¬ `statement`, up to metadata annotations, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A141412.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A141412.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A141412.result' proves 'Math0.Statements.A141412.statement'
```

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [4] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit command), and checks that the pinned dependencies are unmodified. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A141412.lean` in the directory `lean`, once the statement module has been built with `lake build Math0.Statements.OEIS.A141412`; on our machine the check takes about 25 seconds, most of it spent loading Mathlib.

References

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[9] The OEIS Foundation Inc., `https://oeis.org/A048594`, entry by O. Marichev.

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[11] The OEIS Foundation Inc., `https://oeis.org/A132393`, entry by P. Delέham.

[12] The OEIS Foundation Inc., `https://oeis.org/A140749`, entry by P. Curtz (2008).

[13] The OEIS Foundation Inc., `https://oeis.org/A141412`, entry by P. Curtz (2008); conjecture by M. Marcus (2018); revision #32, Nov. 7, 2023.

Produced by Claude (an AI model developed by Anthropic) in the math0 project